HW #4

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1. Consider $K = \mathbb{Q}(\sqrt[3]{28})$. Let $\alpha = \sqrt[3]{28}$, so that the minimal polynomial of α over $\mathbb{Q}$ is $f(X) = X^3 - 28$.

(1) Compute $D(1, \alpha, \alpha^2)$. Deduce that if $p \neq 2, 3, 7$ is a rational prime, then $(p, [\mathcal{O}_K : \mathbb{Z}[\alpha]]) = 1$.

(2) Use the Dedekind's criterion with α to compute the prime ideal factorization of (5) in $\mathcal{O}_K$.

(3) Let
$$\beta = \frac{-\alpha^2 + 2\alpha + 2}{6} \in K.$$
Show that $\beta \in \mathcal{O}_K$ by showing that the minimal polynomial of β over $\mathbb{Q}$ is $g(X) = X^3 - X^2 + 5X + 1$.

(4) Compute $D(1, \beta, \beta^2)$. Deduce that $(3, [\mathcal{O}_K : \mathbb{Z}[\beta]]) = 1$.

Hint. Use that $D(1, \beta, \beta^2) = [\mathcal{O}_K : \mathbb{Z}[\beta]]^2 \operatorname{disc}(K)$.

(5) Use the Dedekind's criterion **with** β to compute the prime ideal factorization of (3) in $\mathcal{O}_K$. What will happen if you mindlessly used the Dedekind's criterion with α to compute a factorization of (3)?

Proof. (1) Note that $D(1, \alpha, \alpha^2) = -27 \cdot 28^2 = -2^4 \cdot 3^3 \cdot 7^2$ by HW2. As the prime factors of $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ are the prime factors of $D(1, \alpha, \alpha^2)$, the result follows.

(2) Note $X^3 - 28 \equiv X^3 - 3 \pmod 5$. Note that $2^3 \equiv 3 \pmod 5$, so $X^3 - 3 \equiv (X-2)(X^2 + 2X + 4) \pmod 5$. Note that $X^2 + 2X + 4 = (X+1)^2 + 3$, and since $\left(\frac{-3}{5}\right) = \left(\frac{2}{5}\right) = -1$, $X^2 + 2X + 4$ is irreducible in $\mathbb{F}_5[X]$. Thus, $X^3 - 3 = (X-2)(X^2 + 2X + 4) \pmod 5$ is the prime factorization in $\mathbb{F}_5[X]$. Thus, by Dedekind's criterion,
$$(5) = (5, \alpha - 2)(5, \alpha^2 + 2\alpha + 4).$$

(3) Note that
$$\beta^2 = \frac{(-\alpha^2 + 2\alpha + 2)^2}{36} = \frac{\alpha^4 - 4\alpha^3 + 8\alpha + 4}{36} = \frac{28\alpha + 8\alpha - 4 \cdot 27}{36} = \frac{36\alpha - 108}{36} = \alpha - 3,$$
$$\beta^3 = \frac{(\alpha - 3)(-\alpha^2 + 2\alpha + 2)}{6} = \frac{-\alpha^3 + 5\alpha^2 - 4\alpha - 6}{6} = \frac{5\alpha^2 - 4\alpha - 34}{6}.$$
So
$$\beta^3 + 5\beta = \frac{5\alpha^2 - 4\alpha - 34 + 5(-\alpha^2 + 2\alpha + 2)}{6} = \frac{6\alpha - 24}{6} = \alpha - 4 = \beta^2 - 1,$$
so $\beta^3 - \beta^2 + 5\beta + 1 = 0$, as desired.

(4) One way to compute this is that
$$g(X + \frac{1}{3}) = (X + \frac{1}{3})^3 - (X + \frac{1}{3})^2 + 5(X + \frac{1}{3}) + 1 = X^3 + X^2 + \frac{X}{3} + \frac{1}{27} - X^2 - \frac{2}{3}X - \frac{1}{9} + 5X + \frac{5}{3} + 1$$

$$= X^3 + \frac{14}{3}X + \frac{70}{27},$$

which has the same discriminant as the discriminant of $g(X)$, and this can be computed using HW2, as $-27\left(\frac{70}{27}\right)^2 - 4\left(\frac{14}{3}\right)^3 = -\frac{70^2+4\cdot 14^3}{27} = -\frac{2^2\cdot 7^2}{27}(25+8\cdot 7) = -\frac{2^2\cdot 7^2\cdot 81}{27} = -2^2\cdot 3\cdot 7^2$. Since $[\mathcal{O}_K : \mathbb{Z}[\beta]]^2$ divides $D(1,\beta,\beta^2)$, 3 does not divide the index.

(5) We can use $g(X)$ to factorize 3. Note that $X^3 - X^2 + 5X + 1 \equiv X^3 - X^2 - X + 1 = (X-1)(X^2-1) = (X-1)^2(X+1)$, so by Dedekind we have

$$(3) = (3, \beta - 1)^2(3, \beta + 1).$$

Note that $f(X)$ fatorizes mod 3 as $X^3 - 1 = (X-1)^3$, so it sounds like if we just use Dedekind using α then 3 should be totally ramified in K, but in fact it is not.

□

Question 2. Let $K = \mathbb{Q}(\alpha)$ with $\alpha \in \mathcal{O}_K$. Let $f(X) \in \mathbb{Z}[X]$ be the minimal polynomial of α over $\mathbb{Q}$. Suppose that p is a rational prime such that $f(X)$ mod p factors into a product

$$f(X) = f_1(X)\cdots f_r(X) \pmod p,$$

such that $f_1(X), \cdots, f_r(X) \in \mathbb{F}_p[X]$ are mutually distinct monic irreducible polynomials in $\mathbb{F}_p[X]$.

Our goal is to show that, under these assumptions, $(p, [\mathcal{O}_K : \mathbb{Z}[\alpha]]) = 1$.

(1) Suppose on the contrary that p divides $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then, it divides $D(1, \alpha, \cdots, \alpha^{n-1})$, where $n = \deg f$. Recall that

$$D(1, \alpha, \cdots, \alpha^{n-1}) = \prod_{i<j}(\alpha_i - \alpha_j)^2,$$

where $\alpha_1, \cdots, \alpha_n$ are the roots of $f(X)$ in the Galois closure L of $K/\mathbb{Q}$. Deduce that, if p divides $D(1, \alpha, \cdots, \alpha^{n-1})$, then there are $i \neq j$ such that $\alpha_i - \alpha_j \in \mathfrak{p}$ for some prime ideal $\mathfrak{p} \subset \mathcal{O}_L$ lying over p.

(2) Show that $f(X)$, as an element of $(\mathcal{O}_L/\mathfrak{p})[X]$, has repeated roots.

(3) Using that $\mathcal{O}_L/\mathfrak{p}$ is also a finite field, and that $(\mathcal{O}_L/\mathfrak{p})/\mathbb{F}_p$ is a separable extension, show that, if $f(X)$ has repeated roots in $\mathcal{O}_L/\mathfrak{p}$, then its factorization into monic irreducible polynomials in $\mathbb{F}_p[X]$ must have some multiplicities. This gives rise to a contradiction.

(4) Deduce that p is unramified in K.

Proof. (1) Self-explanatory.

(2) Self-explanatory.

(3) By (1) and (2), $f(X) \in (\mathcal{O}_L/\mathfrak{p})[X]$ is divisible by $(X-\alpha)^2$ for some $\alpha \in \mathcal{O}_L/\mathfrak{p}$. Let $p(X) \in \mathbb{F}_p[X]$ be the minimal polynomial of α over $\mathbb{F}_p$. Then $p(X)$ is separable, so this implies that $p(X)^2$ divides $f(X)$ (as $f(X) \in \mathbb{F}_p[X]$). Contradiction.

(4) Since $(p, [\mathcal{O}_K : \mathbb{Z}[\alpha]]) = 1$, we can use Dedekind's criterion, and by the factorization, we see that p is unramified.

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